

BE10X AI HACKATHON

College ID Card Portal

Offline-First · AI-Assisted · Zero Infrastructure

A single HTML file that automates the entire student ID card pipeline — from data collection to print-ready PDF — with no internet required.

THE PROBLEM

2–3

Weeks

average wait for
ID cards after joining

100%

Manual

Canva / Photoshop
for every card

0

Integration

between data collection
and card generation

📄 Paper forms or Google Forms — no direct link to card generation

✂️ Manual design in Canva / Photoshop for every single student

🌐 Systems collapse during orientation when campus networks are overloaded

✖️ Transcription errors from re-typing data into design software

THE SOLUTION

One HTML file. No internet. No server. No installation.

Student Portal

Login · Batch select · Fill form
Upload photo · Live ID card preview

Admin Dashboard

Add students individually or bulk
import via CSV · Stats & filters

PDF Generation

CR80-sized ID cards with photo,
QR code, and college branding

Print Ready

One-click print formatted for A4
Bulk export all cards at once

HOW IT WORKS — User Flow



Student Side

- PRN login → batch selection screen
- Form: name, DOB, blood group, mobile, email, address
- Photo upload with 2MB validation
- Real-time completion progress bar
- Live card preview before download

Admin Side

- Add students or bulk CSV import
- Filter by batch · stats dashboard
- Generate all ID cards with one click
- Export CSV + photo ZIP download
- Reset passwords · edit college info

TECH STACK & ARCHITECTURE

pdf-lib

PDF card generation — embedded, no CDN

Vanilla JS

Zero frameworks — no build step needed

JSZip

Bulk photo ZIP export — fully offline

localStorage

All data on-device — total privacy

QRCode.js

QR code with student JSON payload

Claude (AI)

Architecture, debugging & code generation

⚡ Key Decision: Replaced jsPDF with pdf-lib mid-build when emoji rendered as black boxes — StandardFonts solved it completely.

IMPACT & BENEFITS

10 min

vs 3 weeks

Total time from
data to printed card

₹0

Infrastructure Cost

No server, cloud,
or subscriptions

100%

Offline

Works without internet
on orientation day

✓ **Zero Errors**

Data flows from form directly to card — no manual re-typing

🔒 **Full Privacy**

All student data stays on-device in localStorage

⚡ **Instant Setup**

Open one HTML file — no install, no IT team required

📁 **Bulk Operations**

Generate all cards, export CSV + photo ZIP in one click

FUTURE SCOPE



IndexedDB upgrade for larger photo storage (beyond 5MB limit)



College ERP integration via REST API for automatic student sync



Progressive Web App (PWA) — installable, offline-capable on mobile



Analytics dashboard — batch-wise completion, submission tracking



Encrypted JSON backup & restore for multi-device data portability



AI face validation on uploaded photos (orientation, clarity check)